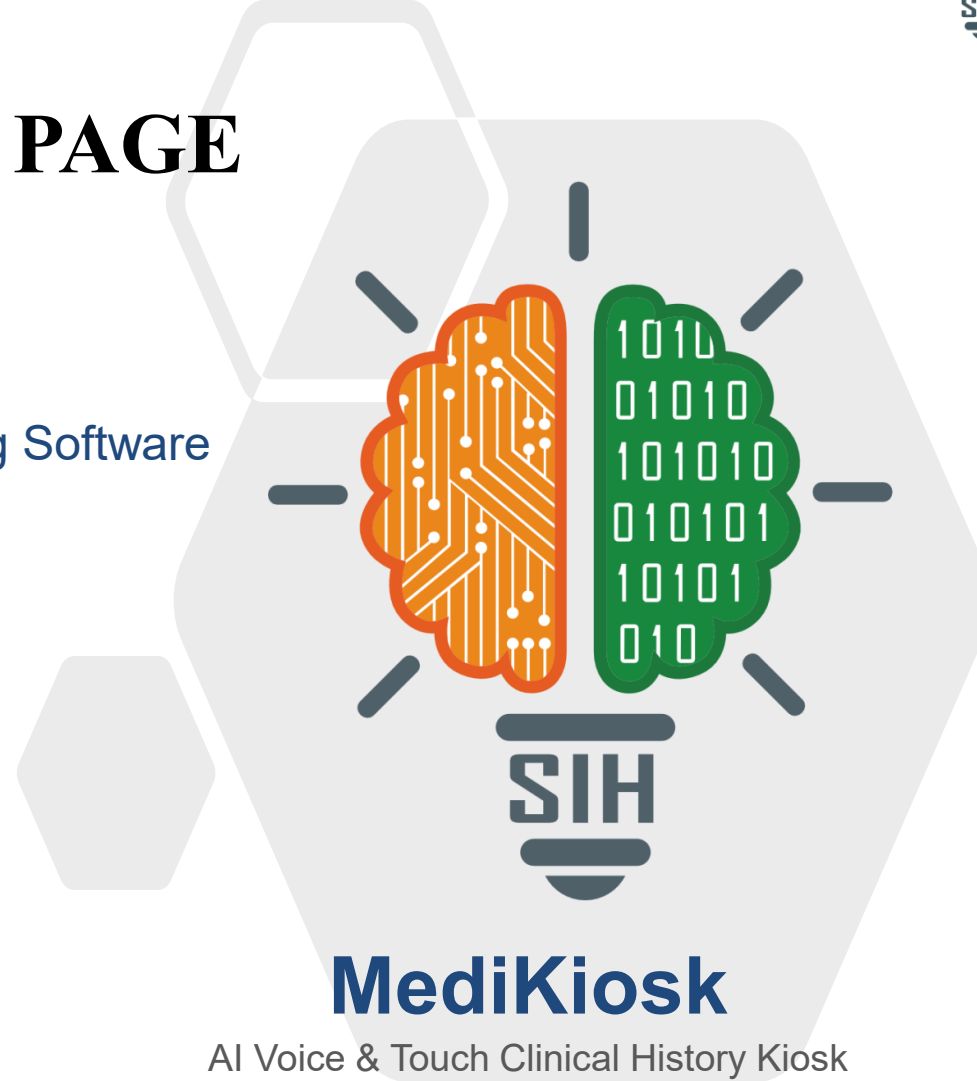


TITLE PAGE

- **Problem Statement ID** – 26047
- **Problem Statement Title** – Patient Case-Taking Software
- **Theme** – MedTech / BioTech / HealthTech
- **PS Category** – Software
- **Team ID** – 138632
- **Team Name** – CODEBLOOM 2.0
- **Team Leader Name** – Shashi Shekhar Rai
- **College Name** – Galgotias University

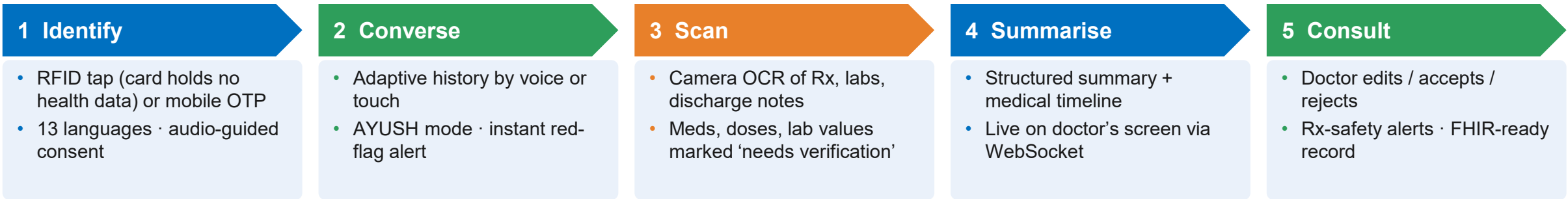


IDEA TITLE

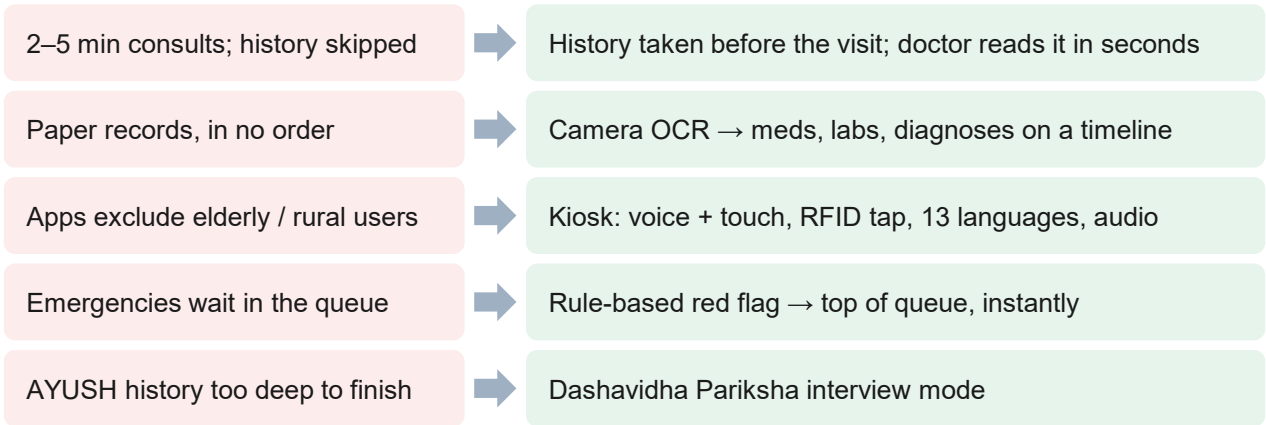
MediKiosk – AI Voice & Touch Clinical History Kiosk with Medical Document OCR

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

■ Detailed explanation of the proposed solution



■ How it addresses the problem



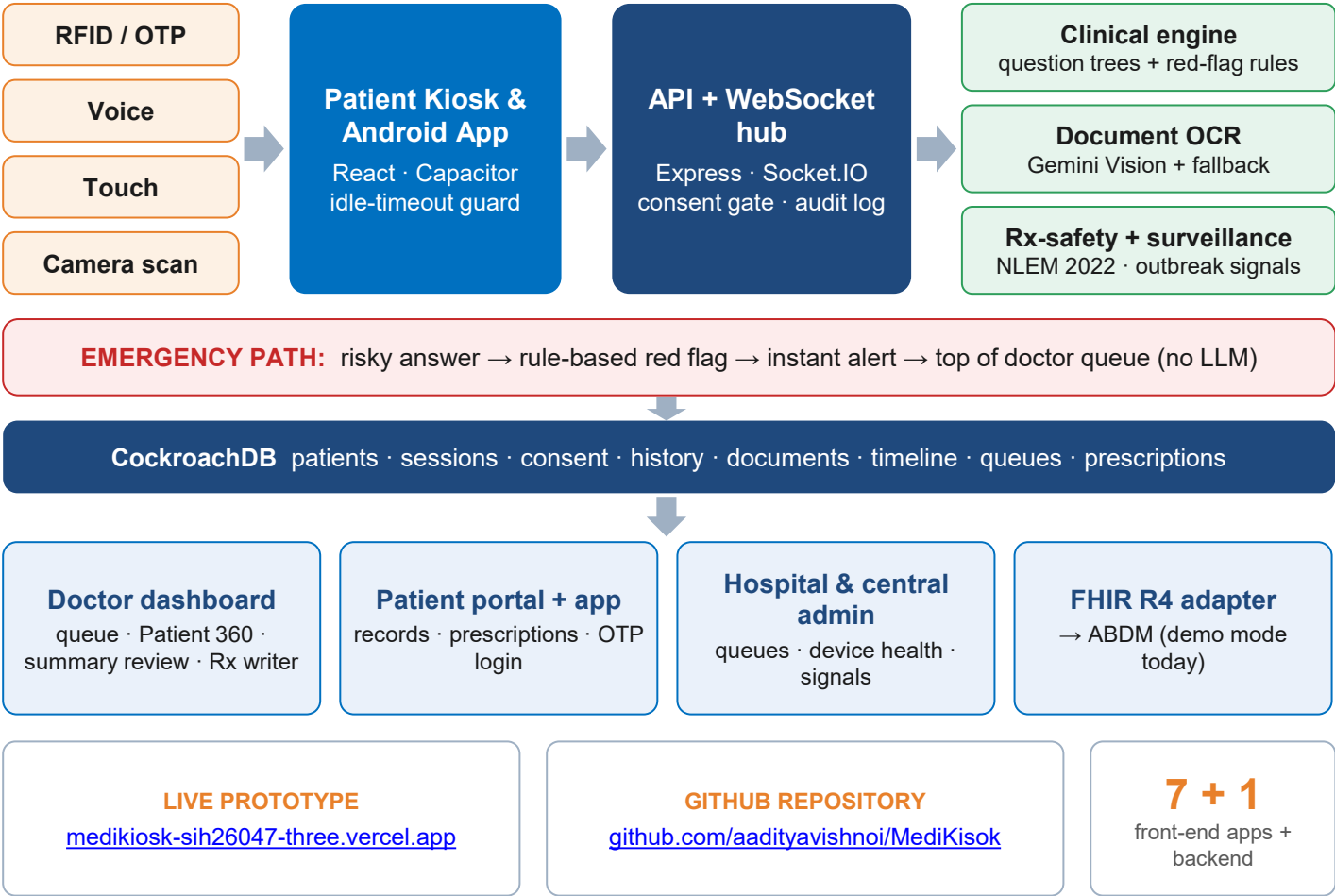
■ Innovation and uniqueness of the solution

- Patient-driven history + document digitisation in ONE flow – beyond admin check-in
- Safety path is rule-based: no LLM decides emergencies
- Every question answerable by voice OR touch – zero training
- Allopathic and AYUSH question trees in one engine
- Privacy-by-design: RFID card stores only a random ID; consent gates all data
- Same data feeds Rx-safety (NLEM 2022, Jan Aushadhi) and regional surveillance

Technologies to be used (e.g. programming languages, frameworks, hardware)

Frontend	React 18, TypeScript, Vite, Tailwind CSS; Capacitor for the Android patient app
Backend	Node.js, Express, Socket.IO (real-time alerts), Prisma ORM, Zod validation
Database	CockroachDB (PostgreSQL-compatible); multi-hospital, consent-aware schema
Clinical AI	JSON clinical question trees + deterministic red-flag rules; NLEM-based Rx-safety engine
Speech & OCR	Web Speech API for voice in/out (Bhashini planned); Gemini Vision OCR with safe fallback
Hardware	Arduino Nano + MFRC522 RFID reader (USB serial); touchscreen PC / tablet; camera
Standards	FHIR R4 (ABDM / NRCeS profiles, demo mode); REST + WebSocket APIs

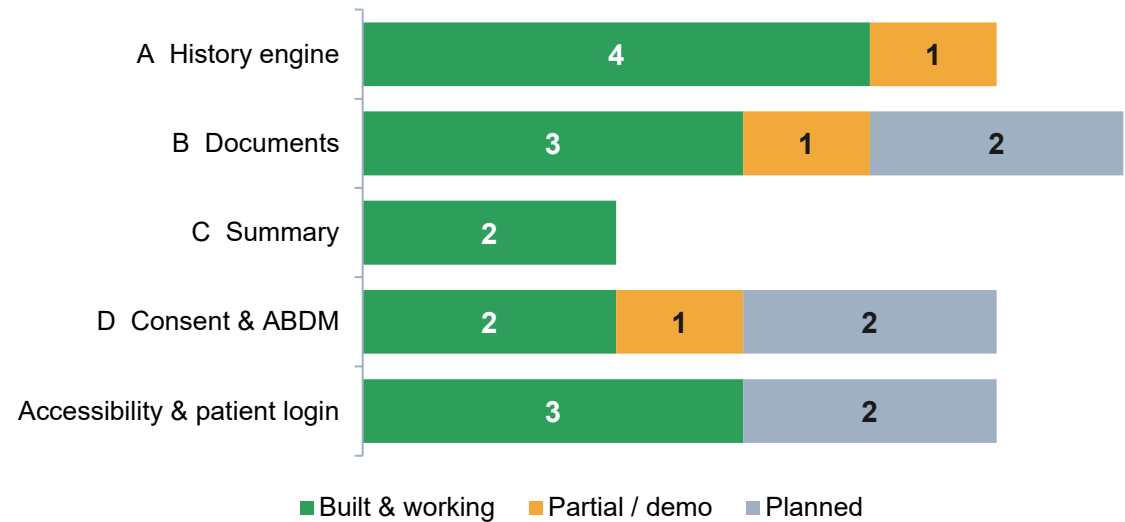
Methodology and process for implementation (Flow Charts/Images/ working prototype)



Analysis of the feasibility of the idea

TECHNICAL	OPERATIONAL	ECONOMIC
- Prototype already runs end to end (7 front-end apps + real-time backend) - Mature open stack; speech, OCR and safety engines are swappable	- Fits beside existing registration / HIS via REST + FHIR - Touch fallback; zero-training use in a waiting room	- Off-the-shelf hardware: PC / tablet, camera, RFID module - Frees doctor time; suggests Jan Aushadhi generics

Problem-statement requirements vs. working prototype (23 items, our own code audit)



Potential challenges and risks	Strategies for overcoming these challenges
Speech accuracy in noisy wards / accents	Touch is always available; Bhashini / AI4Bharat models via pluggable speech interface
OCR errors on handwriting	Confidence scores, ‘needs verification’ status, original image shown to doctor
Data protection (DPDP Act 2023)	Consent-first flow, audit log, no health data on RFID card, role-based doctor login, idle timeout
Clinical correctness of questions	Clinician-reviewed trees; doctor confirms every summary; AI never diagnoses
Adoption & connectivity	Commodity hardware, touch fallback, REST/FHIR fit beside existing HIS; offline mode on roadmap

Future scope (phased rollout)

A – Strengthen	B – Integrate	C – Pilot	D – Scale
- Bhashini / AI4Bharat speech 13-language question banks - OCR benchmark on handwriting - Real BP / SpO2 vitals pod	- Live ABDM / ABHA + FHIR bundles - HIS / EMR connectors - Revocable consent - Admin-portal login (RBAC)	- OPD pilot with clinicians - Time-motion study - Clinical validation of question trees	- Sign-language avatar - Offline-first mode - District outbreak heat-maps - State-level rollout

4,000–10,000

OPD patients per day in tertiary hospitals

2–5 min

typical doctor time per patient

70–80%

of diagnoses possible from a good history

13

Indian languages in the patient interface

■ Potential impact on the target audience

- P

Patients
Shorter waits; history recorded in their own language; records kept digitally
- D

Doctors
Complete structured history on arrival; more time to examine and counsel
- N

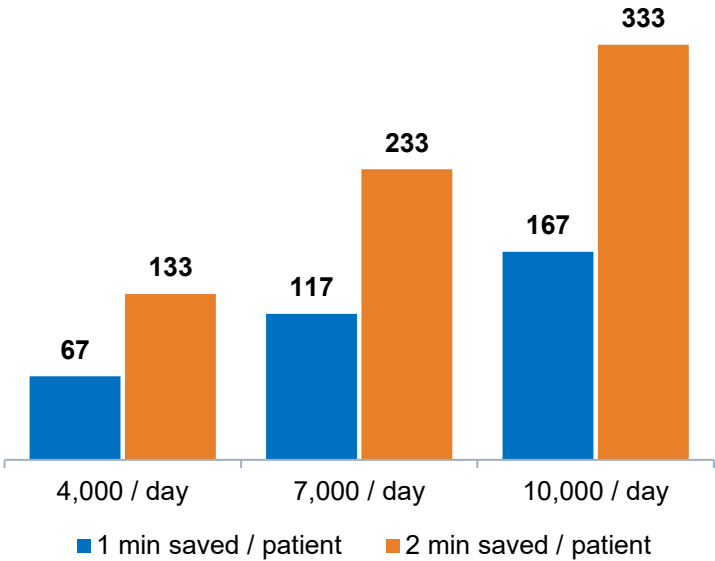
Nurses & admins
Less registration / transcription work; emergencies flagged in real time
- A

AYUSH vaidyas
Dashavidha Pariksha captured without eating into consult time
- H

Health system
Structured, consented data for ABDM and public-health surveillance

Projected doctor-hours freed per day

= patients per day × minutes saved ÷ 60. Assumption: 1–2 min of history-taking saved per patient (to be validated in pilot)



■ Benefits of the solution (social, economic, environmental, etc.)

SOCIAL

- Inclusive for elderly, rural and low-literacy patients
- Earlier detection of emergencies
- Same access in 13 languages

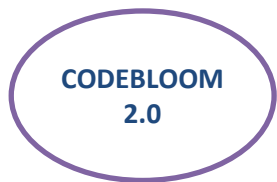
ECONOMIC

- Doctor-hours freed (see chart)
- Low-cost commodity hardware
- Cheaper Jan Aushadhi generics suggested

ENVIRONMENTAL

- Paperless intake and digital records
- Less paper for forms and repeated history

Sources: SIH 2026 problem statement 26047 (volumes, consultation time, 70–80% figure); doctor-hours chart is simple arithmetic on those volumes, not a measured result.



RESEARCH AND REFERENCES



■ Details / Links of the reference and research work

LIVE DEPLOYMENT

medikiosk-sih26047-three.vercel.app

Apps: /kiosk /doctor /portal /app /rfid /hospital /central

GITHUB REPOSITORY

github.com/aadityavishnoi/MediKisok

Source code, model cards and data-source notes

#	Reference / source	Link	How we used it
1	SIH 2026 Problem Statement 26047 – Patient Case-Taking Software (Ministry of Ayush / All India Institute of Ayurveda)	sih.gov.in	Problem scope, four modules, patient journey
2	Irving G, et al. International variations in primary care physician consultation time: a systematic review of 67 countries. BMJ Open 2017;7:e017902	doi.org/10.1136/bmjopen-2017-017902	Evidence behind the short-consultation problem
3	Ayushman Bharat Digital Mission – National Health Authority (ABHA, Health Information Exchange)	abdm.gov.in	ABHA ID field; consent and interoperability roadmap
4	National Resource Centre for EHR Standards – ABDM FHIR R4 profiles	nrces.in	Patient / Encounter resource mapping
5	Digital Personal Data Protection Act, 2023 (MeitY)	meity.gov.in – DPDP Act 2023	Consent-first design, data minimisation
6	Bhashini and AI4Bharat – Indian-language speech (ASR / TTS) models	bhashini.gov.in · ai4bharat.org	Planned upgrade of the pluggable speech interface
7	National List of Essential Medicines 2022 (MoHFW); PM Bharatiya Janaushadhi Pariyojana; Integrated Disease Surveillance Programme (NCDC)	mohfw.gov.in · janaushadhi.gov.in · idsp.mohfw.gov.in	Rx-safety catalogue, generic alternatives, surveillance data model
8	W3C Web Speech API; Arduino MFRC522 library; Google Gemini API	wicg.github.io/speech-api · github.com/OSSLibraries/Arduino_MFRC522	Voice I/O, RFID reader firmware, vision OCR

Prototype status figures on slide 4 come from our own code audit. Volumes and consultation-time figures come from the problem statement and the BMJ Open review above.